

Thermal Reserve

Coordinated thermostat setbacks that keep Southcentral Alaska's gas demand under supply on the coldest days of a multi-day cold snap.

Join from your phone: thermal-reserve.vercel.app/home · MHacks 2026 · Team: [H1 NAME], [H2 NAME], [H3 NAME] UNFINISHED



THE PROBLEM

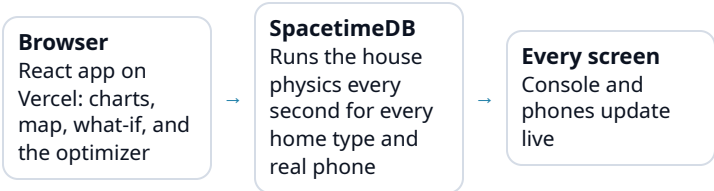
- Enstar's president warned lawmakers the utility could enter this winter about **3 Bcf short** of gas, the equivalent of **18 days without heat** in sustained cold.¹
- On Aug 7, 2026, Cook Inlet storage held about **6.54 Bcf** of its 13 Bcf working capacity.¹
- In a shortage, Enstar would ask **large commercial and industrial customers to cut back first**, before homes, and imported LNG may not arrive until **late 2029**.²
- Thermostat programs elsewhere save gas during short events, but homes reheat afterward ("snapback"). In ConEd's pilot, snapback erased **52%** of the savings.³

WHAT WE BUILT

A **simulation**, not control of real thermostats. An operator console plans setbacks for a simulated fleet of Anchorage homes (24 home types, shown as 1,000 dots on a map). An optimizer picks the smallest total temperature drop that keeps each day's demand under supply, planning across the whole cold snap so reheating lands on days with spare gas.

Households join from a phone, see their indoor temperature and gas saved, and can opt out with one tap. The server never lets a home drop below **62°F**, and homes that need steady heat are exempt.

HOW IT WORKS



Browser and server run the same physics file; a test checks they agree.

What it doesn't do: close the seasonal shortfall. Over 20 cold days, 25,000 homes save about 28 MMcf, under 1% of 3 Bcf. Thermal Reserve helps on the coldest days, when businesses would otherwise be cut.

HONEST IMPACT

A sustained setback at -20°F outside, automated homes only. derived
Each home's heat loss is calibrated from Enstar's 149 Mcf average home,⁴ 30 years of Anchorage weather,⁵ and Alaska's gas energy content (988 BTU per cubic foot).⁶

Enrolled homes	Gas saved, 5°F setback	Share of Enstar's 20 MMcf/day peak supply ⁷	Value at \$17.50/Mcf ¹
10,000	0.57 MMcf/day	2.8%	~\$10,000/day
25,000	1.42 MMcf/day	7.1%	~\$24,900/day
50,000	2.85 MMcf/day	14.2%	~\$49,800/day

Peak supply: extra gas Enstar's supply contract lets it buy on the coldest winter days. \$17.50/Mcf: the price producer HEX offered Enstar for extra gas in 2026.

Two assumptions, checked against data assumed

- 75% of a home's gas goes to space heating.** EIA's 2020 household survey puts Alaska at 72–74%.⁹ At 72%, every savings figure above would be about 4% lower.
- Furnace efficiency 0.85.** Gas furnaces must be at least 80% efficient (federal minimum since 2015; 95% from Dec 2028),¹⁰ and Anchorage homes mix older and newer units. It doesn't change the gas numbers: we calibrate from gas actually burned, so efficiency cancels out.

DOES THE MODEL MATCH REAL PILOTS?

We ran the same physics under each pilot's conditions. derived

Test	Published	Ours	Pass band	Result
Savings kept after reheating (ConEd-like)	48% ³	46.5%	38–58%	Pass
Net daily savings (SoCalGas-like)	2.2% ⁸	1.14%	1.5–3.0%	Gap

Pass bands are our own choice, set around each published value before testing. We show the gap instead of tuning it away: the two pilots imply different amounts of snapback, and on daily savings our model errs low.

SOURCES (EACH LINK OPENED AND CHECKED OCT 3, 2026) · EVERY NUMBER IS LABELED SOURCED, DERIVED, OR ASSUMED

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2. Homer News, "Winter gas shortfall comes into sharper focus for Southcentral Alaska," Sep 10, 2026. homernews.com/2026/09/10/winter-gas-shortfall-comes-into-sharper-focus-for-southcentral-alaska

3. Con Edison, "Gas Demand Response Report on Pilot Performance – 2019/2020," Jul 1, 2020. documents.dps.ny.gov/DocRefId/7FA119BD-B04D-4378-95A3-BB26697815AC

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7. Petroleum News, "New Enstar supplies," Mar 6, 2016. petroleumnews.com/pntruncate/978589605.shtml

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10. U.S. DOE, consumer furnace standards final rule, Federal Register, Dec 18, 2023. govinfo.gov/content/pkg/FR-2023-12-18/html/2023-25514.htm